

Marcel ASSIE

GEOMATICS ENGINEER

☎ +33 7 68 06 25 78 | ✉ marcelassie2k@gmail.com | 🔗 LinkedIn | 🐙 GitHub | 📁 Portfolio | 📍 Champs-sur-Marne (77)

Geomatics engineering student specializing in spatial data science, passionate about the Geo-Data ecosystem and geospatial technologies. Seeking a **permanent full-time position** to leverage my skills in **geospatial software development**, **data pipeline automation**, and **spatial analysis**, serving high-impact projects that transform complex territorial data into effective decision-making tools.

EDUCATION

École Nationale des Sciences Géographiques (Geodata Paris) Champs-sur-Marne, France
Geomatics Engineering Degree • Spatial Data Science Specialization Sept. 2023 – Sept. 2026

Institut National Polytechnique Félix Houphouët-Boigny Yamoussoukro, Ivory Coast
Bachelor's Degree in Surveying Oct. 2019 – Jul. 2022

Lycée Scientifique de Yamoussoukro Yamoussoukro, Ivory Coast
French Baccalauréat (Scientific Stream) Sept. 2016 – Jul. 2019

PROFESSIONAL EXPERIENCE

ISOGEO Paris, France
GIS Developer – Apprenticeship Sept. 2025 – Sept. 2026

- Led the technological overhaul of "Scan Isogeo" by replacing the FME ETL with new automated processing pipelines.
- Developed extraction and processing modules for various RDBMS (Oracle, SQL Server) and modern geographic formats (GeoPackage, GeoParquet, CAD files) leveraging GDAL and SQL.
- Explored and implemented AI models to automatically extract and enrich semantic metadata.
- Drafted functional and technical specifications, and contributed to project management within an Agile team.

ORYJIN Lille, France
Geospatial Data Scientist – Internship Apr. – Aug. 2025

- Researched, audited, and integrated various geospatial and socio-demographic data sources (INSEE, IGN) into the open-source H3 global grid system.
- Developed ETL pipelines and implemented statistical algorithms to calculate geographic aggregates (Python, Snowflake, SQL).
- Extracted spatial information (housing areas, infrastructure) from satellite imagery using Computer Vision AI models.
- Automated data visualization via generative AI: created descriptive models and prompt workflows to generate realistic images describing territories.

IGN (National Institute of Geographic and Forest Information) Forcalquier, France
Geomatics Engineer – Internship May – Jul. 2024

- Conducted applied mini-projects in surveying, photogrammetry, metrology (bridge monitoring), remote sensing, and geodesy, including methodological formalization and uncertainty analysis.
- Designed an automated workflow for mapping traffic restrictions (Alpes-de-Haute-Provence):
 - Structured and cleaned multi-source datasets.
 - Modeled geospatial processes (QGIS Model Builder).
 - Developed quality control and data validation algorithms.

SKILLS

Programming Languages: Python, SQL, JavaScript, HTML/CSS, R, Cypher
Databases: PostgreSQL/PostGIS, Oracle Spatial, SQL Server, Neo4j
Web Technologies & Ops: Docker, GitHub, Django, Flask, Vue.js, React Native
GIS Libraries: GDAL, Proj, GeoPandas, ArcPy, PyQGIS, Shapely, Leaflet.js
Tools & Formats: QGIS, ArcGIS Pro, FME, GeoServer, AutoCAD | GeoParquet, GeoPackage, CAD
Soft Skills: High adaptability, Rigor, Critical and analytical thinking, Proactive

PROJECTS

- **Contextual Data Extraction (SDGs):**  GitHub
 - Developed an advanced extraction method based on Large Language Models (LLMs) and structured it into knowledge graphs to analyze relationships between Sustainable Development Goals (PyTorch, Transformers, NLP, LLMs, Neo4j).
- **Mobile Navigation App (French National Gendarmerie):**  GitHub
 - Developed a Waze-inspired application specifically tailored for public security missions (React Native, Flask).
- **Spatial Statistics Study (Île-de-France):**  GitHub
 - Modeled the impact of transport network changes on commuting modal shifts through a spatio-temporal panel approach and prospective scenario simulation (Scikit-Learn, Seaborn, SciPy, Proj, GeoPandas).
- **KlassIvoire SaaS Platform:**  Website
 - Contributed to the development of a digital school monitoring platform, integrating AI tools for performance analysis.
 - Technologies: Django, SQL, PostgreSQL, Docker, Cloudflare R2, Redis, Gemini API.
- **Educational Institution Analysis Platform:**  GitHub
 - Designed an intelligent geographic monitoring platform to leverage spatial data for decision-making.
 - Technologies: Django, PostgreSQL/PostGIS, Leaflet.js.

EXTRACURRICULAR ACTIVITIES & VOLUNTEERING

Private Tutor	Île-de-France
<i>In-home Tutoring – Mathematics and Physics</i>	<i>Since Oct. 2023</i>
Provided tailored pedagogy and methodology, highly attentive to students' needs.	
VERTIGEO – ENSG Geomatics Junior Enterprise	Champs-sur-Marne
<i>General Secretary</i>	<i>Feb. 2024 – Mar. 2025</i>
Drafted meeting minutes, managed archives and administrative partnerships, and organized meetings.	
Épicerie Solidaire Campésienne (Solidarity Grocery)	Champs-sur-Marne
<i>Volunteer</i>	<i>Feb. 2024</i>
Collected food and hygiene products at a Carrefour store. Promoted community exchange and open-mindedness.	

LANGUAGES & INTERESTS

Languages: French (Native), English (Professional Working Proficiency - TOEIC B2)
Interests: Artificial Intelligence, Tech Watch, Soccer, Acrobatic Gymnastics, Board Games